

FRANCOIS SANDY LEUTOU

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EDUCATION

Brown University — Providence, RI

Expected May 2029

Sc.B Applied Math-Computer Science

- **Relevant Coursework:** CSCI0190: Accelerated Intro to CS, CSCI0300: Fundamentals of Computer Systems, CSCI1970KF: Agentic Studio, APMA1655: Intro to Probability & Statistics

GBHS Nkol-Eton — Yaounde, Cameroon

June 2023

- High School Diploma(GCE A-Level) | June 2023s

EXPERIENCE

Software Engineering Intern | Amea | Yaounde, CMR (Hybrid) | *May 2025 – August 2025*

- Collaborate cross-functionally on development, release, and maintenance of production software for a growing startup, participating in weekly sprint planning and progress reviews

Program Facilitator & SAT Tutor | Open Dreams | Yaounde, CMR | *Sep 2024 – Aug 2025*

- Facilitated SAT prep sessions for 30+ students, driving score improvements of 100+ points per student through targeted instruction and personalized feedback
- Strengthened public speaking and instructional design skills by leading structured group sessions

Co-Founder & Head Administrator | Grade Masters Tutoring Community | Yaounde, CMR | *Jan 2024 – May 2025*

- Founded and scaled tutoring community to 20+ students; organized mentor-tutor pairings and directly mentored 2 students, helping 4 achieve 3.5+ GPAs
- Taught Math and Computer Science (~10 hrs/week), designing curriculum to target individual student weaknesses

CS Teaching Assistant (A & O-Level) | GBHS Nkol-Eton | Yaounde, CMR | *Oct 2021 – Jun 2023*

- Led revision sessions and supervised programming labs for A/O-Level CS students, contributing to 100% A-Level pass rate across cohort
- Assisted with grading, administrative work, and curriculum support (~5 hrs/week)

CS Club President | GBHS Nkol-Eton | Yaounde, CMR | *Oct 2022 – Apr 2023*

- Led 13-member club; taught C programming fundamentals to juniors and mentored peers through end-of-year CS projects, strengthening team's technical foundations

PROJECTS

Image Classification Web App | Python, React, FastAPI, PyTorch | [Github](#) | [Live Site](#) | *April 2026 - May 2026*

- Built an end-to-end image classification system using a pretrained CNN (ResNet50) in PyTorch to predict object labels from user-uploaded images
- Developed a RESTful API using FastAPI to handle image uploads, preprocessing, and real-time inference
- Implemented image preprocessing pipeline (resizing, normalization, tensor conversion) to ensure compatibility with model input requirements
- Designed a lightweight frontend interface to upload images and display top-3 predictions with confidence scores
- Architected a client-server system enabling scalable model inference via HTTP requests

LeetCode Progress Tracker (Full-Stack Web App) | Python, React, SQLAlchemy, FastAPI | [GitHub](#) | [Live Site](#) | *May 2026*

- Designed and implemented a full-stack web application to track and analyze coding problem progress with CRUD functionality
- Built RESTful FastAPI backend to manage problem metadata including difficulty, topics, and notes
- Modeled relational database schema (PostgreSQL) to support efficient querying and filtering by topic and difficulty
- Leveraged AI-assisted development tools to accelerate frontend prototyping and feature iteration
- Integrated progress tracking features (e.g., problems solved over time, distribution by difficulty) to provide actionable insights
- Developed React frontend enabling sub-second filtering and analytics across tracked problems
- Deployed a production-style full-stack architecture integrating Vercel-hosted frontend with railway-hosted backend and PostgreSQL database
- Working on multi-user support, providing the resource for multiple users to track their progress

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks: React, FastAPI, Flask, Tailwind CSS, SQLAlchemy

Developer Tools: Git, Docker, Bash, Linux, AI-assisted development tools (Claude Code, GitHub Copilot)

Languages (Human): French (Native), English (Fluent), Japanese (Beginner)

Interests: Hiking, psychology, soccer, chess, reading, pool